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Citizens' Satisfaction with the Land Registration Procedure: Evidence from Ha Dong District, Hanoi, Vietnam

Quang Cuong Doan ^{a, b*}

^a *Department of Urban Planning and Design, The University of Hong Kong*

^b *University of Science, Vietnam National University, Hanoi, Vietnam*

Abstract: This study aimed to discover citizens' evaluation and satisfaction with land registration procedures. Using Hanoi as a case study, interviews were done at the Land Registration Office to clarify the satisfaction with land registration. The results show that the satisfaction rate was 74.6%. Collected data were analyzed through machine learning algorithms to find the factors affecting satisfaction. The results indicate that the number of times citizens go to agencies and waiting time are the most significant factors contributing to citizens' satisfaction. The government should design and unify single points for applications, develop a digital-based cadastral database, enable online payment, and modernize facilities to improve the citizens' satisfaction.

Keywords: citizens' satisfaction, land registration, land management, Hanoi, Vietnam.

1. Introduction

A cadastre was defined as “a parcel-based and updating land information system which contains land record of interest” (FIG, 1995). It describes a parcel's geometry that can be linked with other records (FIG, 1995; Kha & Vo, 2007). It has been considered fundamental to socio-economic and environmental sustainability (Rajabifard et al., 2014). The cadastral template was launched in 2003 to develop a generic template for

countries. With this template, countries can describe cadastral status, undertake land administration, and improve demand (Daniel Steudler et al.). In the land management context, the cadastre is understood as the science of state land management (Kha & Vo, 2007), describing the land management function to include the land registration system, cadastral mapping, land information system, land use planning, land valuation and legal information (Williamson, 2000).

Land registration (LR) and issuance of land use right certificates (LURC) are ways to guarantee land users' rights (Abdulai & Owusu-Ansah, 2014; Place & Migot-Adholla, 1998). LR protects people's land ownership and the legitimate interests of land users. At the same time, it supervises how people use land and safeguards the common interests of the whole society. LR is also considered an important factor in developing countries' economies (Gershon Feder & Nishio, 1998). It helps decision-makers in these countries manage land resources effectively. LR creates a premise for implementing state management tasks such as land use planning, land lease, land classification, land allocation, valuation, inspection, and resolution of land disputes.

In Vietnam, LR is part of the 15 state land management functions (The National Assembly, 2013). Vietnamese land registration is based on the title system. This system uses civil legal documents to confirm the legal status of parcels and owners in form of a unified certificate. The land registration system includes land registry, cadastral record establishment, and issuance of LURC. There are two types of land registration: first-time registration and changing registration. While the first-time registration applies for uncertified or new parcels, the changing registration is carried out for parcels with certificates but with changes in land use status, land users' information, or when land users use their land rights (transaction, inheritance, mortgage, lease, etc.). In Vietnam, the State only assigns land use rights to land users. Land users have rights and responsibilities

for performing their obligations under the law. Through the LURC, land registration stipulates the legal responsibilities between the State and land users in observing land laws. Cadastral records and LURC provide complete information and a solid legal basis for determining land use rights. It acts as a tool to protect land users' rights when disputed or infringed. It also defines financial obligations on land use and the effective use of land. However, the land registration system has limitations that affect its efficacy.

The issuance of certificates is carried out slowly, causing difficulties for land management and adversely affecting socio-economic development. The delay in the land registration process also affects people's confidence in the government apparatus. Over the years, many scientists have conducted land registration studies in various aspects, with few focusing on citizens' satisfaction. The studies have focused on the economic impact of land registration on society (Griffith-Charles & Opadeyi, 2009), the rental market (Qin et al., 2020), the benefits of land registration (Ali et al., 2019; G. Feder, 1987), effective land registration in housing projects (Barry & Whittal, 2016), unofficial land registration (Lai et al., 2016), land registration services transformation (Cuong, Tuan, & Hung, 2019), land registration with the cadastral system (Alkan, Gürsoy Sürmeneli, & Polat, 2021; Gürsoy Sürmeneli & Alkan, 2021), land registration evolution (Divithure & Tang, 2012), the efficacy of the cadastral system (Getie, Birhanu, & Dadi, 2022). In 2008, (Deininger et al., 2008) analyzed the initial impact of rural land certification in Ethiopia. These results contribute to making recommendations to ensure sustainability and improve initial land registration efficiency. (Alban Singirankabo & Willem Ertsen, 2020) explored land registration's effect on the relationship between land tenure security and agricultural productivity. The author has argued that the difference between systematic and sporadic registration and registration ways can affect agricultural productivity. (Ojo, 2014) investigated user satisfaction by using frequency distribution,

proportion and weighted mean score. (Babatunde et al., 2014) explored high costs, costs incurred by officials, and delayed processes are obstacles to land registration.

These studies mentioned above have significantly contributed to the theoretical foundation for understanding land registration and land use rights. However, studies on citizens' satisfaction with land registration are rare in the literature. Since citizens' satisfaction is essential in assessing the land administration system performance (Daniel Steudler, Rajabifard, & Williamson, 2004). This study aims to answer the following questions: (1) What is the level of citizens' satisfaction with land registration in Ha Dong district, Hanoi, Vietnam? (2) How to determine factors that affect citizens' satisfaction? And (3) What factors affect citizens' satisfaction with the land registration procedure?

2.Literature review

2.1. Regulation related to land use rights and land registration in Vietnam.

From 1945 to 1979, no legal documents were issued on land registration and LURC. Therefore, the land use status was carried out by rapid surveys to record land information. The documents during this period were mainly cadastral maps (measured manually by tape measure) and catalog books. Documents were in paper form and could not be queried for land-use history. From 1980 to 1988, land registration was done through government legal documents. During this time, the State issued decision 201/CP stipulating unified land management, which required rigorously reviewing the process. However, the record systems at that stage mainly reflected the land use status. With the Land law, which was first issued in 1987, the General Department of Land Administration (GDLA) decided to issue LURC. Since 1990, land registration has been widely implemented, but the number of certificates issued is small. The implementation of the land law in 1993 led to the subsequent recognition of land as valuable property. The State then allowed land transfer,

posing an urgent need for certification.

In 2001, the National Assembly approved the amendment of the Land law 1993 and issued replacement documents. After ten years of implementing the Land law 1993, a new Land Law was promulgated in 2003. It introduced regulations on land use rights certificates and guidance on cadastral records management. It is worth noting that there have been additions to the regulatory framework. Circular No. 29/2004 regulates the cadastral database - this was Vietnam's first regulation for computerizing the land registration system. Decree 88/2009 and Circular 07/2009 created a significant turning point in reforming administrative procedures in land registration. The government also prescribed a uniform certificate template nationwide for land and houses on the land.

Vietnam was able to improve land governance in 9/63 provinces through the Vietnam Land Administration Project (VLAP) from 2009 to 2013, with the assistance of the World Bank. VLAP aimed to develop a sustainable land database that served multi-purpose. VLAP helped the government simplify land registration procedures, enhancing the quality of services and people's awareness of land services. This project achieved initial success. The proportion of people satisfied with land management in 9 provinces increased from 58% in 2013 to 78% in 2015 (The World Bank, 2015). Digital cadastral maps have also been made in some districts of these provinces. However, the project only covered nine provinces. Public services on land achieved a basic level (people still take procedures in person and unavailable online payment). The cadastral maps are in paper form in many other provinces.

In 2013, a new Land law was passed to help perfect the contents related to land registration. In particular, the National Assembly stipulated the land information system as one of the 15 contents of land management. This contributed to the modernization of the land registration system. Besides this, there are many other legal documents related

to the land registration database, such as Circular No. 75/2015 of the Ministry of Natural Resources and Environment (MoNRE) about technical standards on the land database; and Circular No. 05/2017/TT-BTNMT of the MoNRE, regulating the building of land databases. In 2016, the MoNRE approved the Project "Strengthening land management and land database" in Decision No. 1236/QĐ-BTNMT to build a multi-purpose land information system at the national level. This system was based on system architecture, synchronous infrastructure, and unified software across the country, ensuring the operation of the database system of all 63 provinces. The project also aimed to complete and operate the national land database (cadastral data, land-use planning data, land price data, and land inventory). The project has the benefit of providing public services and sharing land information with relevant sectors (tax, notary, banking). In addition, it supports strengthening land-use management. It ensures the consistent implementation of the Land law 2013, establishing a system to monitor and evaluate land use management and improving public services in the land sector.

Table 1 near here

From table 1, the legal system of land registration in Vietnam is still complicated and has yet to achieve stability. Specifically, the Land law changes frequently (every 5 to 10 years). When the new Land law was issued, it led to a change in the guiding documents. Since the introduction of the Land Law, regulations on cadastral records have been issued five times in decision No.499, Circular No.1990/2001, Circular No.29/2004, Circular No.09/2007, and circular No.24/2014. However, some policies need to be aligned with reality. For example, Circular 29/2004 and Circular 09/2007 stipulate how to record land use purposes in the cadastral books. From 2004 to 2008, no new regulations guided how to record land information on cadastral maps. This led to a situation where the old regulations were used in some provinces, and new regulations were applied in

other places. Until 2013, the new Land law mentioned constructing a land information system. Vietnam issued new regulations to address conflicts and disputes and improve the land management system. However, a new land law may also result in new challenges. This led to the land law in Vietnam constantly changing to adapt to the new situation. However, it also causes many difficulties in the implementation process.

2.2. Land registration procedure in Vietnam

The land registration procedure in Vietnam begins with preparing applications for registration and culminates in issuing LURC. Initially, the land users submit documents (dossier) to the Commune People's Committee (CPC) or the land registration office (LRO). The dossier includes the application form, papers proving the land use right, house (if any), diagram of housing, construction works, proof of financial obligations, identity card, and household registration. Then, documents will be checked at the CPC by cadastral staff. The CPC is responsible for certifying the land-use status, the land origin, and conformity with the land-use planning and disrupted status. However, many records cannot verify the land origin because of lacking documents proving the land origin. The CPC must organize a consultation with the residential community to solve this problem. This makes the whole process time longer. Then, the CPC must notify the LRO to measure the parcels if the area does not have a cadastral map. The measuring time shall not exceed ten working days and shall not include in the total time.

The results of the dossier checking process must be made public within 15 days. Dossiers that pass the checking process are valid and proceed to the next step at the LRO. In the case of invalid dossiers, the land users are notified to complete the information for validation. The settlement time of the CPC shall not exceed 15 working days after receiving the valid dossier. The LRO conducts verification, verifying the documents and confirming the applications. Cases that are not eligible are returned to the land users. For

cases eligible for certificate issuance, the LRO shall extract the cadastral map of the parcels. The LRO shall send cadastral data to the tax office for cases that must fulfill financial obligations. The tax authority determines the financial obligation level, then sends a notice of payment to the LRO. The LRO shall forward the message of invoices to the land users. The land users are responsible for paying the money at the treasury or the tax office to get payment proofs. Land users submit proof of payment at the Office of District People Committee. The LRO makes a list of the certificates for the eligible cases. After that, dossiers are checked and appraised by The Natural Resources and Environment Agency (NREA). Ineligible cases are returned to the LRO.

The final step is certificate issuance. The People's Committee of the district, and province level, together with the NREA, considers and signs decisions on granting the certificates and transfers the certificates to the LRO. Users then collect the certificate from the initial place of application.

Figure 1 near here

3. Study area and methodology

3.1. Study area

Ha Dong district is a critical gateway of the Hanoi capital. It is about 13 km from the Hanoi center to the southwest. Ha Dong's natural area is 4,833.66 hectares. As the center of culture, economy, science, technology, and education of the former Ha Tay province, the Ha Dong district is situated in the urban chain of the Northern critical economic region. It creates advantages in socio-economic development. Ha Dong district experienced rapid urbanization, especially when Ha Dong was merged into Hanoi city in 2008. Demand for land and housing growth led to the number of real estate transactions increasing rapidly. Ha Dong has the most exciting land market in the Southwest region,

Hanoi. This has led to a vast number of land transactions and registrations. The number of applications for certification has increased rapidly (from about 15,000 in 2019 to 18,000 in 2020). However, the cadastral records, such as inventory and land change monitoring books, have not been implemented synchronously on the land information system. Hence, the rapid urbanization has exposed challenges in land registration.

Figure 2 near here

3.2. Data collection and methodology

a, Data collection

The author collected data through interviews with citizens to answer the research questions. The interviews were conducted at the LRO in March 2021. At this time, 138 citizens were randomly chosen to evaluate satisfaction with the land registration procedure. The questionnaire included individual information and citizens' opinions about land registration procedures. Questions were designed based on administrative procedure evaluation index regulations such as total time, the number of times to contact the office, publicizing managerial procedures, and staff attitude (Government, 2018). The Likert scale was used to measure participants' evaluations. Questions about satisfaction, times, facilities, the complexity of the process, and staff attitude were measured by 5 points Likert (very dissatisfied = 1 and very satisfied = 5). Data were cleaned to remove illogical responses and prepared for analysis.

b, Support vector machine (SVM)

The paper uses the SVM for analysis. SVM is a machine technique that is used for classification and regression. It can split into a predefined number of classes based on a hyperplane with input data and training algorithms. This technique has been used in many

studies, such as in coating classification by (Alshari & Gawali, 2021; Rana & Venkata Suryanarayana, 2020), in the medical field by (Khairandish et al., 2021; Kumar et al., 2021), in the classification of fuels by (Tuttle, Blackburn, & Powell, 2020), in surface temperature prediction by (Zhou et al., 2020). However, the number of studies that used this technique in classification, predicting citizens' satisfaction with land registration, is rare. The advantage of SVM is that it uses numerous samples to create models with good performance, even for small, multi-dimensional (Mountrakis, Im, & Ogole, 2011).

Given is a training dataset (a_i, b_i) , $i=1, \dots, n$, $x \in \mathbb{R}^d$, $y \in \mathbb{R}$ indicates the label; with n being the dimensionality of the dataset, and N is the number of samples. In this paper, the input is seven variables, the labels are the satisfaction of citizens (0: unsatisfied; 1: satisfied). The formula describes the regression result about satisfaction: $f(x) = \text{sign} [\sum y_i \gamma_i K(x_i, x_j) + b]$ where b is the offset value, γ_i is the Lagrange, and $K(x_i, x_j)$ is the kernel function. In this paper, SVM was chosen for evaluating citizens' satisfaction using RStudio programming.

c, Random Forest

The random forest (RF) technique has been widely used in recent years, introduced by Breiman in 2001. RF is highly effective in classifying, regression, and assessing important factors. It aggregates decisions from decision trees to make classifications or predictions. RF divides the dataset randomly to create training and testing datasets. Since its introduction, this technique has been applied in many fields due to its good processing speed and excellent classification (Du et al., 2015). Some areas that have applied this technique with excellent results include remote sensing (Cavallaro et al., 2015; Douna et al., 2021), risk warning (Chen et al., 2021), prediction soil's PH (Makungwe et al., 2021), biomass estimation (Mutanga, Adam, & Cho, 2012), land cover prediction (Ebrahimi et al., 2021), land use fragmentation (Talukdar et al., 2021), impact assessment on

agricultural land (Nabiollahi et al., 2021). However, applications of RF for evaluation of satisfaction in land registration are rare.

d, Model performance evaluation

This study randomly divided a set of samples into two groups (70% for training and 30% for testing). The performances of citizens' satisfaction models are evaluated based on parameters including true satisfaction (TS), true dissatisfaction (TDS), fail satisfaction (FS), and fail dissatisfaction (FDS). In addition, overall accuracy (OA), sensitivity (SENS), specificity (SPEC), and positive predictive ability (PPA) were also calculated to evaluate the model's accuracy. The model with the higher index is the better model. In machine learning, Kappa statistics use as supporting performance comparisons. The model with a higher Kappa index shows lower error costs and more realistic performance (Ben-David, 2008). Finally, the important level of each factor is exported (based on MeanDecreaseGini indexes) to find which ones affect citizens' satisfaction.

4. Results

4.1. Citizen's satisfaction on land registration

From table 2, the result shows that granting certificates has partly met the needs of people carrying out current land registration procedures. Almost everyone went to LRO to register for residential land (108/138 samples). The interview also indicates that 82.6% of people went to LRO twice for different administrative procedures. Majority of respondents were doing land change registration, with 123/138 people. Service quality has improved thanks to the one-stop mechanism. However, the number of documents that need to be processed is extensive, leading to an overload for the application reception department. While the percentage of people who did not have to wait to submit

applications is 80.4%, nearly 20% of respondents complain that they must wait a long time for these procedures.

Facilities for service provision are also factors that affect people's satisfaction. Inadequate facilities, cramped working spaces, and long wait times can affect people's satisfaction. About 79% of people gave a positive evaluation of the facilities. The remaining 21 % of people complain that the facilities do not meet their needs. As for time, 54/138 people complain that the time to carry out the procedures is longer than what is stipulated in the regulations. In fact, after applying at the LRO or the CPC, many people still must go to different agencies to complete the land registration process. Around 32.6% of the interviewed people complained that they had to go to state agencies more than three times to complete applications, pay taxes and fees, and get certificates. Statistical results also show that 34.1% of the respondents rated the staff's attitude as not enthusiastic. Around 65.9% of respondents highly appreciated the sincere attitude and detailed and meticulous instructions. Due to the many different administrative agencies and documents to be submitted, 39.1% of respondents assessed that land registration and certification procedures are still complicated. The managerial procedures stop at level 2 (providing forms); people must go directly to the LRO, CPC, and the tax department to pay land use levies or taxes. The overall assessment of people's satisfaction is that about 3/4 of respondents felt satisfied with land-related administrative procedures. However, the remaining 25.4% of people want further improvements in first land registration and land change registration.

Table 1 near here

4.2. Factors affecting citizen's satisfaction

a. Variable test

This study selected eight questions (Q5 to Q12) to explore people's satisfaction. Before using algorithms, the research used Cronbach's alpha to test the variables' reliability. Many authors suggested this method (Nunnally, 1994), (Cuong & Binh, 2020; Hair et al., 2014). According to Hair, if Cronbach's alpha is higher than 0.6, the variable's reliability is accepted. In this study, Cronbach's Alpha of 8 questions was 0.779, indicating a high-reliability rate and showing consistency in the survey scale.

b. Factors affecting citizen's satisfaction

The paper uses SMV and RF algorithms to evaluate the factors affecting people's satisfaction. From the training dataset in table 3, it can be seen that RF's PPA and SVM's PPA are 100%. This shows that the predictive ability of both models for positive is very good. While the NPA rate of the SVM model is 98.5%, the figure for RF is 97.2. These values indicate that the ability to classify non-satisfaction in SVM is better. As for SENS and SPEC index, while the values of RF are 90.9% and 100%, the figures for SVM are 95.8% and 100%. This result shows RF and SVM's ability to classify citizens' satisfaction correctly. In general, both models fit with the training dataset. However, SVM has better performance (OA = 0.989 and Kappa = 0.971) than SVM (OA = 0.978 and Kappa = 0.938). It is noticeable that there was a better performance of the SVM model. Specifically, PPAs in both models are around 84%. It means the ability to predict citizens' satisfaction in the new dataset is 84%. The prediction probabilities of the non-satisfaction in SVM and RF are relatively high, at 97.6% and 93.9%, respectively. The SVM's SENS value is 90.9%, higher than that of RF by 6%. The SPEC values of models are about 94%. The accuracy and Kappa of RF are 0.913% and 0.786, respectively. However, these values in SVM are slightly higher, at 0.934 for the former and 0.826 for the latter. The AUC of SVM is 0.926, compared to 0.892 in RF. Both methods gave excellent results in predicting satisfaction. But the SVM method has slightly better performance.

In addition, factors influencing citizens' satisfaction were ranked by Mean Decrease Accuracy and Mean Decrease Gini indexes. Factors that have higher indexes mean more important factors. The results of the analysis of the importance of the input variables show that: Q10 (number of times to complete the records) has the most significant influence on citizens' satisfaction, followed by Q5 (waiting time) and Q7 (facilities). Three factors with a relatively strong influence on citizens' satisfaction are Q8 (completion time) and Q6 (overcrowded level), Q9 (staff attitude). Q11 (registration procedure) is the factor that has the most negligible impact on land registration satisfaction (Figure 3).

Table 2 near here

Figure 3 near here

5. Discussion

The land-use right certificate is a legal document that protects the rights and responsibilities of land users. It is also considered a legal document that helps people secure to invest in land. However, the land registration procedure is still limited. This study found that the number of times to complete records and waiting time are the most important factors affecting citizens' satisfaction with land registration. So, to reduce the number of visits to administrative offices and waiting time, the place where people submit application documents for land registration needs to be unified. Currently, people submit documents in different places and multiple times, which takes time for verification. Another solution for cutting the number of times to complete records and waiting is building a cadastral database system. The loss and tearing/ damage of archival documents resulted in the inability to verify the origin of the land. Therefore, the cadastral database is a potential solution for searching parcel history because it can update locations by day

(Karnes, 2004). Furthermore, online payment needs to be applied to land registration tax to reduce the number of times people visit administration offices. With encryption technology to ensure transaction safety, online payment brings convenience and saves time. This paper also found that facilities are also a factor related to satisfaction. Citizens complained about the lack of space for waiting and the lack of computers for declaring and looking up information. Thus, there should be more space and more equipment for people to look up information and declare.

The interview results indicated that nearly 40% of the respondents evaluated the time the procedure takes as slower than the regulation stipulates. Besides the complex history of the parcel, the certification mechanism is also the reason behind this problem. According to regulations, NREA is responsible for signing and certifying the certificate. This leads to overcrowding when the number of certificates is vast. As such, this paper proposes that the LROs at the district branch should be allowed to sign the certificate to reduce pressure and overload on NREA.

The result differs from (Ojo, 2014), who evaluated that friendliness of staff and professionalism need to be improved first. There are two main reasons behind this difference. Firstly, the Vietnamese government has issued regulations on public administrative procedures (including surveying and assessing staff attitudes). Secondly, a code of conduct for staff has also been developed and operated in many localities. This has significantly reduced people's complaints about the staff's attitude when handling land registration records.

6. Conclusion

People's satisfaction measures progress in implementing administrative procedures, especially land-related ones. This article identifies people's satisfaction with

land registration procedures at the LRO in Ha Dong, Hanoi, Vietnam. The analysis results show that 25.4% of people are unsatisfied with land registration procedures. The major factors that contribute to dissatisfaction are the number of times people have to go administrative offices to complete the records, waiting time and facilities. Other factors affecting people's satisfaction are the reception staff's overload and attitude. The results also indicate that performance of SMV is better than RF in evaluating citizens' satisfaction. This study argues that it is necessary to designate and unify single point for application to improve citizen satisfaction and the effectiveness of land registration. Besides, building a digital or ICT-based cadastral database system is essential to support parcel history searches. Finally, enable online payment of fees and taxes using e-payment solutions, and upgrade the facilities to improve the user experience.

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